Chapter 4: Functions and Recursion

4.1 Defining and Calling Functions

Functions are reusable blocks of code that perform a specific task. They help in breaking down complex programs into smaller, manageable sections, improving code readability and reusability. Python's support for functions makes it an ideal language for building modular applications.

Basic Structure of a Function in Python:

A function in Python is defined using the `def` keyword, followed by the function name, parentheses `()` which may contain parameters, and a colon `:`. The function body is indented and can contain one or more statements.

Syntax:

```
def function_name(parameters):
    # Function body
    return value  # Optional
```

Example:

```
def greet(name):
```